공학수학(2)

14.7

Triple Integrals in Cylindrical Coordinates

Triple Integrals in Cylindrical Coordinates

In plane geometry the polar coordinate system is used to give a convenient description of certain curves and regions.

Figure 1 enables us to recall the connection between polar and Cartesian coordinates.

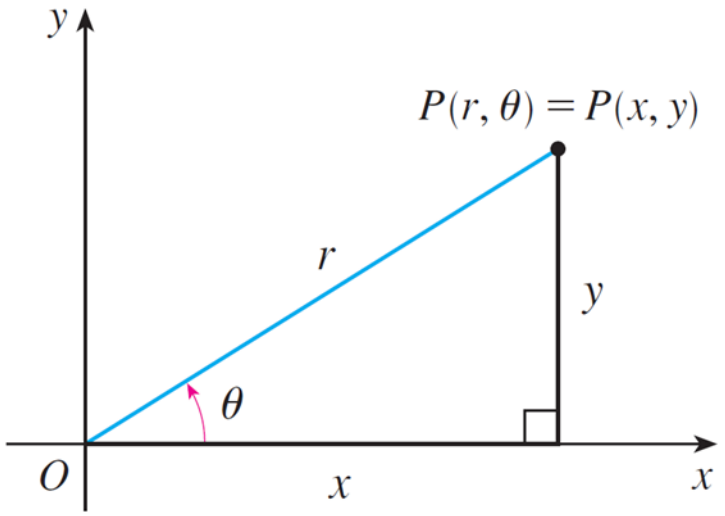


Figure 1

Triple Integrals in Cylindrical Coordinates

If the point P has Cartesian coordinates (x, y) and polar coordinates (r, θ), then, from the figure,

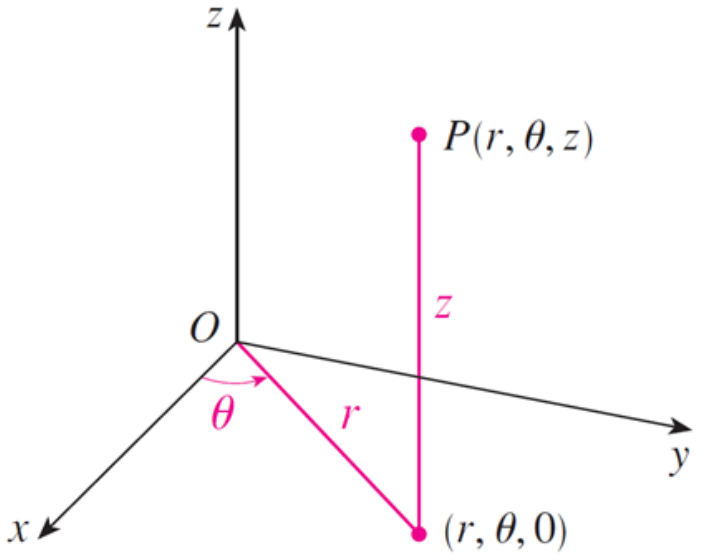
$$x = r\cos\theta \qquad y = r\sin\theta$$

$$r^2 = x^2 + y^2 \qquad \tan\theta \frac{y}{x}$$

In three dimensions there is a coordinate system, called *cylindrical coordinates*, that is similar to polar coordinates and gives convenient descriptions of some commonly occurring surfaces and solids. As we will see, some triple integrals are much easier to evaluate in cylindrical coordinates.

Cylindrical Coordinates

In the **cylindrical coordinate system**, a point P in three-dimensional space is represented by the ordered triple (r, θ, z), where r and θ are polar coordinates of the projection of P onto the xy-plane and z is the directed distance from the xy-plane to P. (See Figure 2.)



The cylindrical coordinates of a point

Figure 2

Cylindrical Coordinates

To convert from cylindrical to rectangular coordinates, we use the equations

$$\textbf{1} \qquad x = r\cos\theta \qquad y = r\sin\theta \qquad z = z$$

whereas to convert from rectangular to cylindrical coordinates, we use

$$\textbf{2} \qquad r^2 = x^2 + y^2 \qquad \tan\theta = \frac{y}{x} \qquad z = z$$

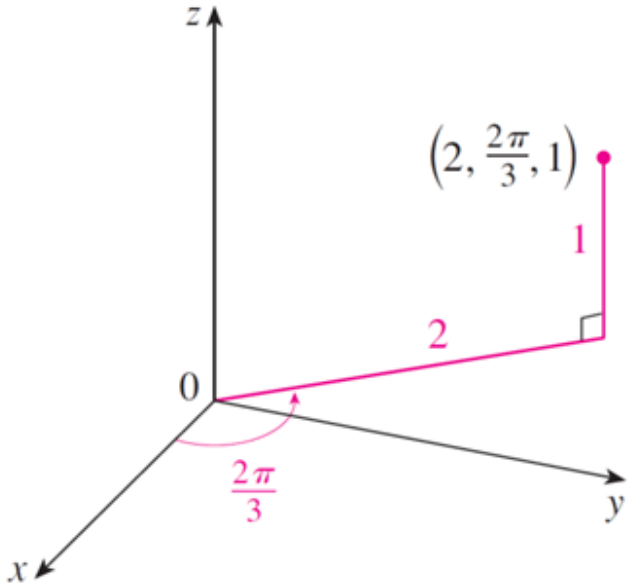
Example 1

(a) Plot the point with cylindrical coordinates $(2,\ 2\pi/3,\ 1)$ and find its rectangular coordinates.

(b) Find cylindrical coordinates of the point with rectangular coordinates $(3, -3, -7)$.

Solution:

(a) The point with cylindrical coordinates $(2,\ 2\pi/3,\ 1)$ is plotted in Figure 3.



Triple Integrals in Cylindrical Coordinates

Suppose that E is a type 1 region whose projection D onto the xy-plane is conveniently described in polar coordinates (see Figure 8).

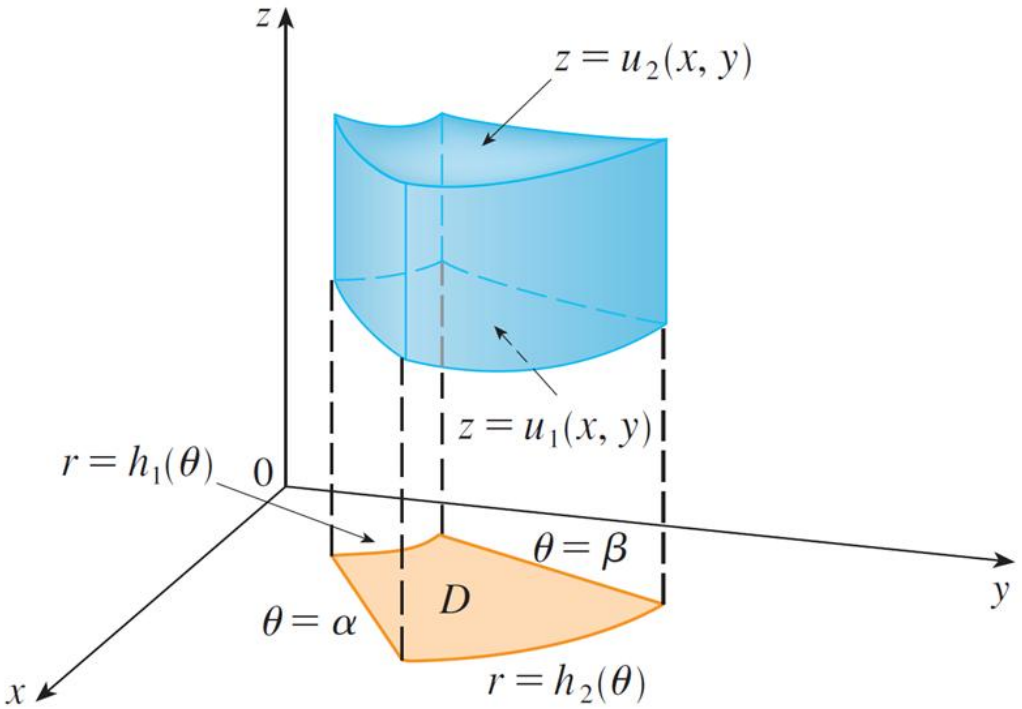


Figure 8

Triple Integrals in Cylindrical Coordinates

In particular, suppose that f is continuous and

$$E = \{(x, y, z) \mid (x, y) \in D,\ u_1(x, y) \le z \le u_2(x, y)\}$$

where D is given in polar coordinates by

$$D = \{(r, \theta) \mid \alpha \le \theta \le \beta,\ h_1(\theta) \le r \le h_2(\theta)\}$$

We know

3
$$\iiint_E f(x, y, z)\, dV = \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z)\, dz \right] dA$$

Triple Integrals in Cylindrical Coordinates

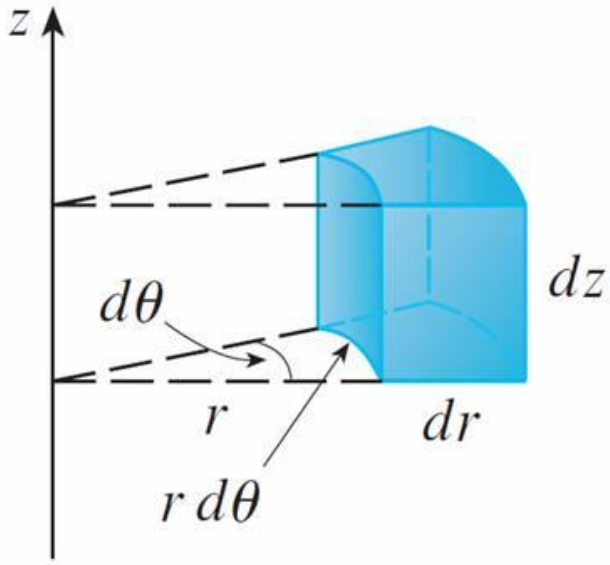
But to evaluate double integrals in polar coordinates, we have the formula

4 $$\iiint_E f(x,\ y,\ z)\ dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r\cos\theta,\ r\sin\theta)}^{u_2(r\cos\theta,\ r\sin\theta)} f(r\cos\theta,\ r\sin\theta,\ z)\ r\ dz\ dr\ d\theta$$

Formula 4 is the **formula for triple integration in cylindrical coordinates**.

Triple Integrals in Cylindrical Coordinates

It says that we convert a triple integral from rectangular to cylindrical coordinates by writing $x = r\cos\theta$, $y = r\sin\theta$, leaving z as it is, using the appropriate limits of integration for z, r, and θ, and replacing dV by $r\,dz\,dr\,d\theta$. (Figure 9 shows how to remember this.)



Volume element in cylindrical coordinates: $dV = r\,dz\,dr\,d\theta$

Figure 9

Triple Integrals in Cylindrical Coordinates

It is worthwhile to use this formula when E is a solid region easily described in cylindrical coordinates, and especially when the function $f(x, y, z)$ involves the expression $x^2 + y^2$.

Example 5

Evaluate

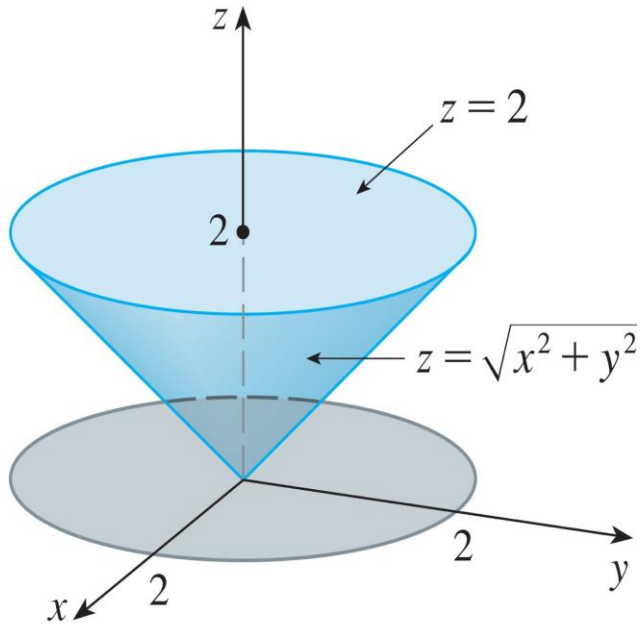


FIGURE 9